

MINI USB - MICRO USB TO USB-C ADAPTER



PRODUCT

[HTTPS://SHOP.GILTESA.COM/PRODUCT/MINIUSB-TO-USB-C-ADAPTER](https://shop.giltesa.com/product/miniusb-to-usb-c-adapter)

[HTTPS://SHOP.GILTESA.COM/PRODUCT/MICRO-USB-TO-USB-C-ADAPTER](https://shop.giltesa.com/product/micro-usb-to-usb-c-adapter)

PLEASE READ THROUGH THESE INSTRUCTIONS ENTIRELY BEFORE ATTEMPTING TO INSTALL

WARNING: IF YOU ARE NOT COMFORTABLE WITH SOLDERING, OR PERFORMING ANY STEP IN THIS GUIDE, DO NOT PERFORM THE INSTALL YOURSELF. FIND SOMEONE WHO IS COMFORTABLE TO DO IT FOR YOU.

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DESCRIPTION

The **MiniUSB** and **MicroUSB to USB-C** are adapter boards that allows you to replace a MiniUSB or MicroUSB connector with a modern and standard USB-C port.

These boards are for general use and can be used in a wide variety of devices, allowing you to replace the old connector with the new USB-C.

This board has been successfully tested on the following devices, which are documented in this installation guide:

MiniUSB:

- RCA adapter to HDMI

MicroUSB:

- Amazon Kindle Paperwhite 3 (7th Gen. June 30, 2015)
- BlitzWolf BW-LT5 Pro IP68 Waterproof LED lamp.

FEATURES

- Replace the MiniUSB or MicroUSB connector with a USB-C, with data connection supported.
- Charge your device with:
 - USB power banks
 - USB-A chargers
 - USB-C chargers
 - USB-C PD chargers (normal speed, not fast)
 - USB-A to USB-C cables
 - USB-C to USB-C cables

INCLUDED

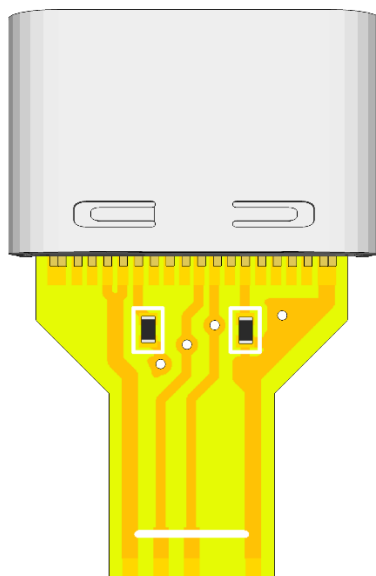
- 1 flexible board.

RECOMMENDED / REQUIRED [NOT INCLUDED]

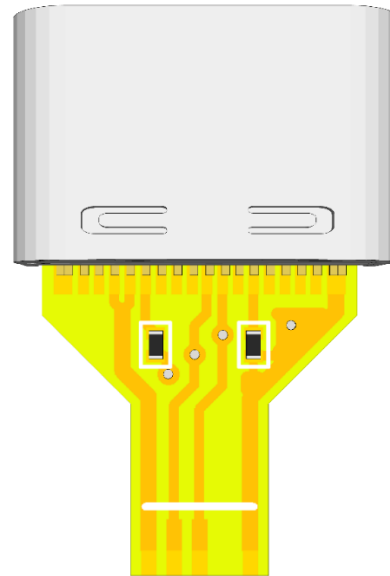
- Screwdriver.
- Soldering iron / hot-air soldering station.
- Kapton tape.
- Tin.
- Flux.
- Desoldering pump.
- Desoldering mesh.
- Tweezers.
- Scissors.
- Cutting pliers.
- Isopropyl alcohol.

BOARD DETAILS

These small flexible boards have a total of 4 pads.



MiniUSB



MicroUSB

The pads, from left to right, correspond to:

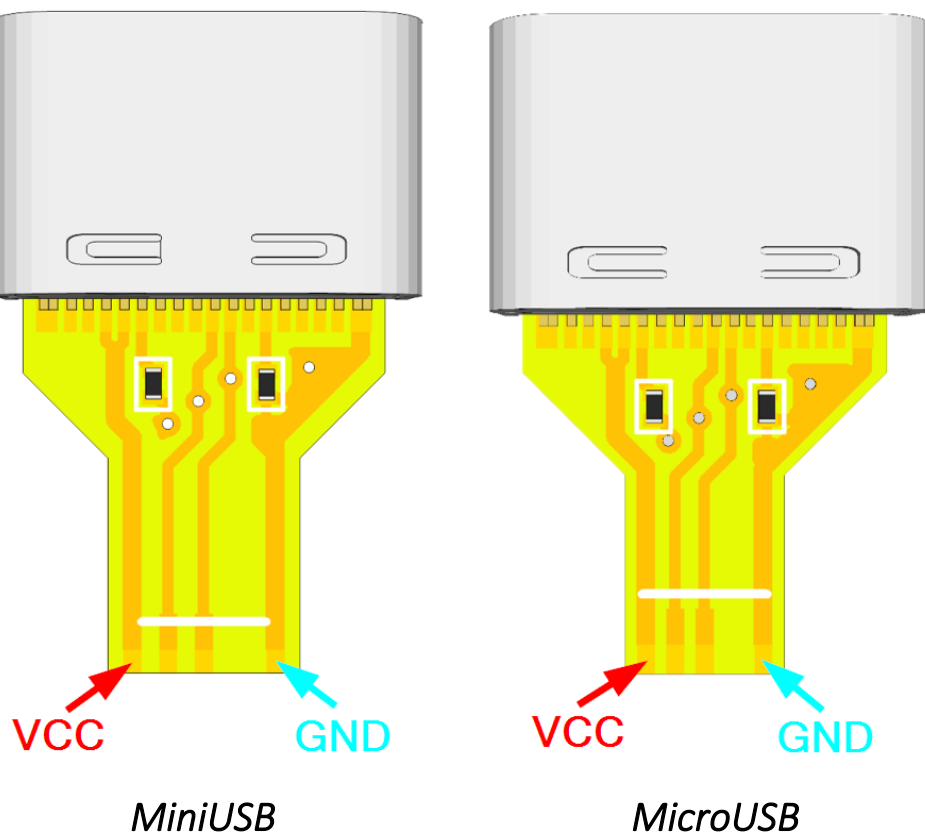
1. **VCC** The +5V pad.
2. **D-** Negative pad of the data connection.
3. **D+** Positive pad of the data connection.
4. **GND** The ground pad.

IMPORTANT NOTE: There are MiniUSB and MicroUSB connectors with reversed polarity. Make sure your device uses the standard polarity before using these adapters. *(This is explained in detail in the first chapter of the installation guide)*

TEST THE BOARD!

Before starting the installation, you should test the board. If it doesn't work contact me [for a replacement](#) (*all boards are fully tested, but they may damage during the shipping, we try to package them as better as possible*), if it works, go ahead with the installation.

Connect the power from your USB charger to the USB-C connector on the board. Then, with a multimeter in voltage measurement mode, **check for a 5V reading**. If that's the case, continue with the installation.



Unfortunately, it is not possible to test the data connection until the board is fully installed.

INSTALLATION STEPS

Since this product is intended for general use with devices that have a MiniUSB or MicroUSB connector, it's not possible to provide step-by-step instructions for each specific device.

However, the procedure is similar, and even if your device is not the one shown below, the steps explained will help you with the installation.

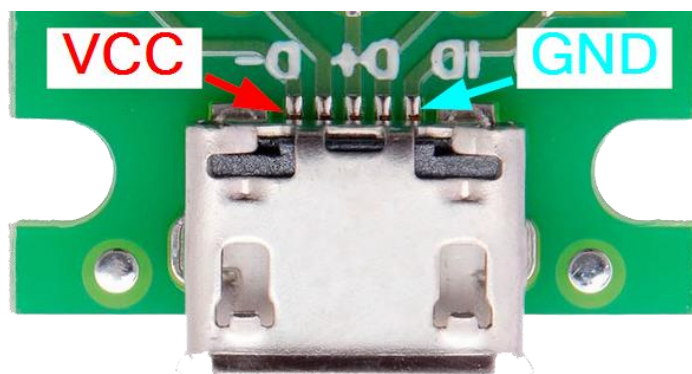
DETERMINE IF YOUR DEVICE IS COMPATIBLE WITH THIS BOARD

First, even before purchasing this board, you should disassemble your device to check several key points that will determine whether the use of this board is possible or not.

1. POLARITY OF YOUR MINIUSB OR MICRO USB CONNECTOR

The MiniUSB / MicroUSB connectors comes in several models/designs, depending on the needs of the manufacturer who designed your device. The most suitable MiniUSB / MicroUSB version may have been used. The key point is to **check the polarity** of the connector, as **it could be either standard polarity or reverse polarity**.

This board is designed for devices with standard polarity. This means that with the MiniUSB / MicroUSB connector facing you, the **VCC pad should be on the left** and the **GND pad on the right**.



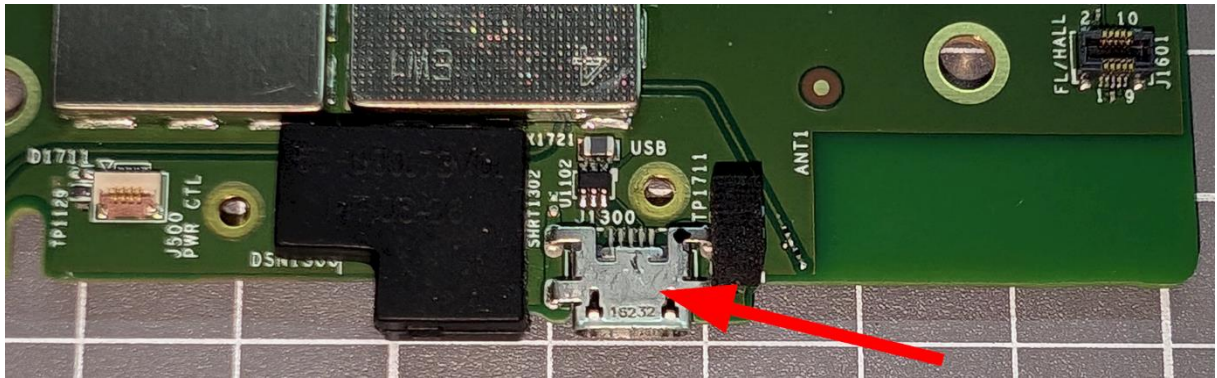
INSTALLATION ON AMAZON KINDLE PAPERWHITE 3 [MICRO USB]

Disassemble your device carefully; you can use the [iFixit guide](#) for more details. Be careful not to lose any screws⁽¹⁾. It is recommended to take a photo before and after removing each part, so you can follow the steps in reverse when reassembling it.

⁽¹⁾ The screw next to the MicroUSB connector cannot be installed once the assembly is complete, as the flexible circuit is located in that exact spot.

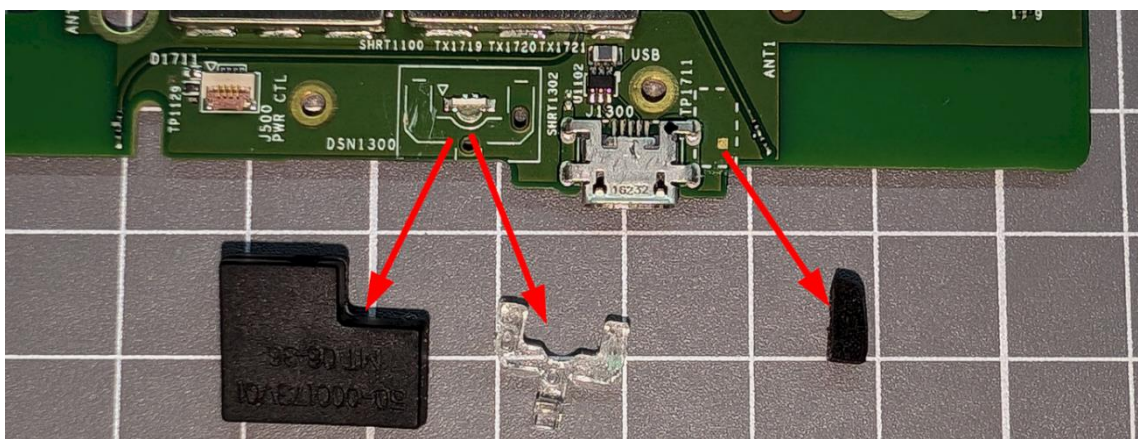
1. REMOVE THE MICRO USB CONNECTOR

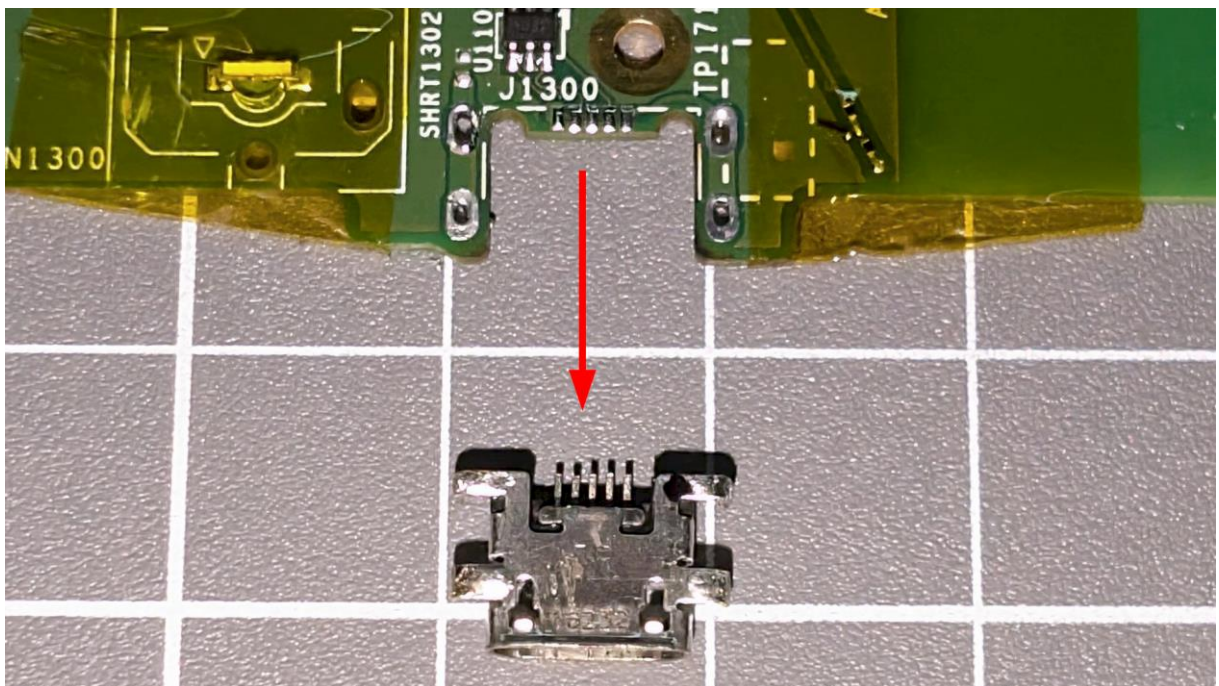
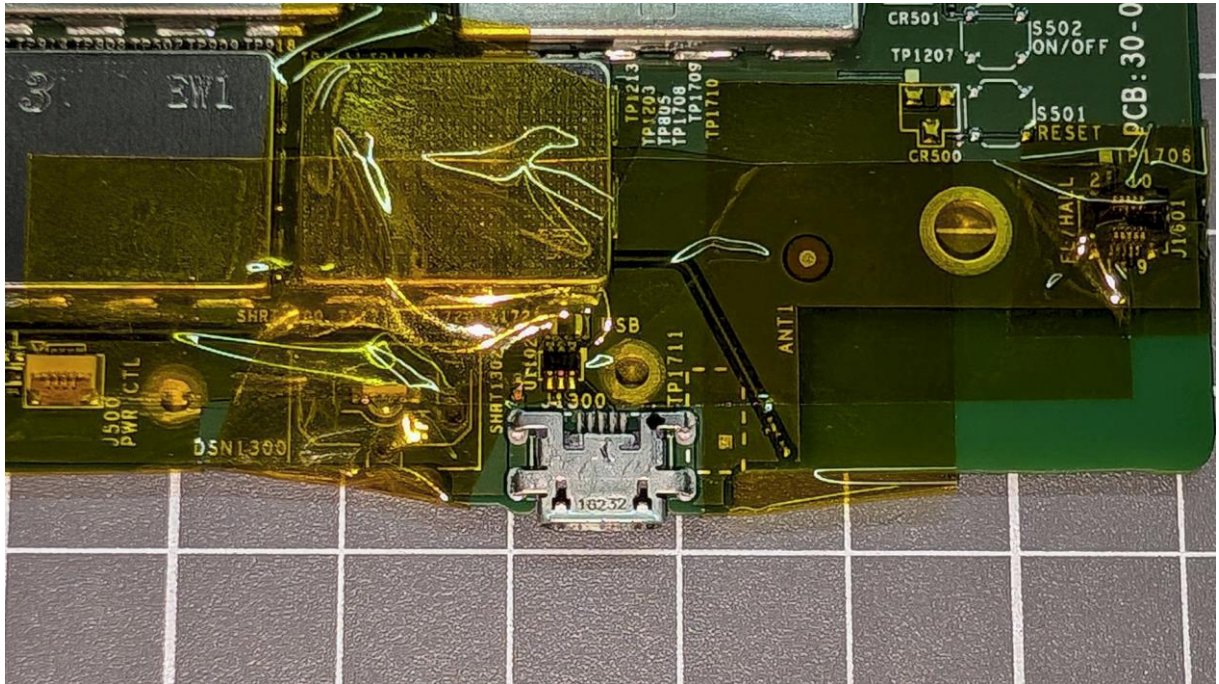
It is only necessary to remove the original power connector from the mainboard.



You can remove it using a hot air rework station (don't forget to protect nearby areas, especially the plastic parts, with kapton tape), or you can use a soldering iron and a desoldering pump.

The ground plane is quite large, so desoldering the connector requires a fair amount of work. Keep that in mind. *(This means that when you apply heat with the soldering iron, it dissipates quickly, preventing the solder from melting)*



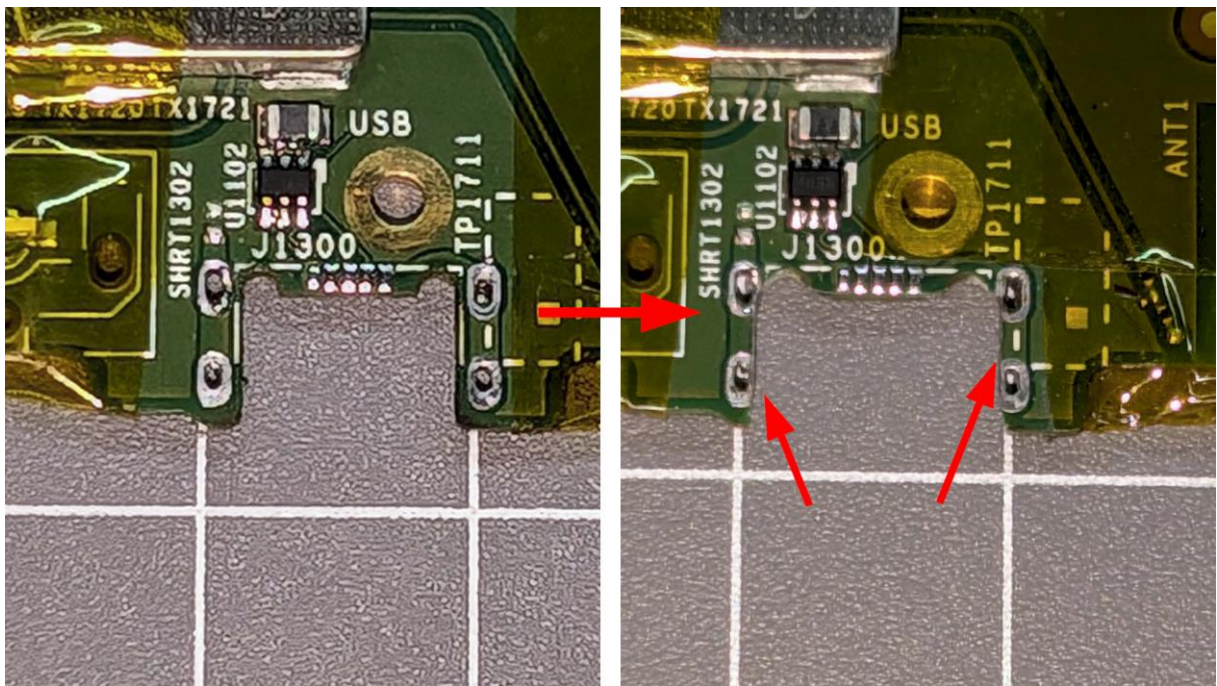


Once removed, make sure there are no solder residues in any of the holes. Clean the board with isopropyl alcohol.

2. FILE THE MAINBOARD TO MAKE SPACE

For this device, where the MicroUSB connector is placed in a slot and soldered at each end instead of underneath, it's necessary to file the mainboard on both sides until the width of the slot is large enough for the USB-C connector to fit.

Use small metal files to file the PCB. You need to file it down almost to the pad holes. In any case, file the board gradually and test repeatedly until the connector fits properly.

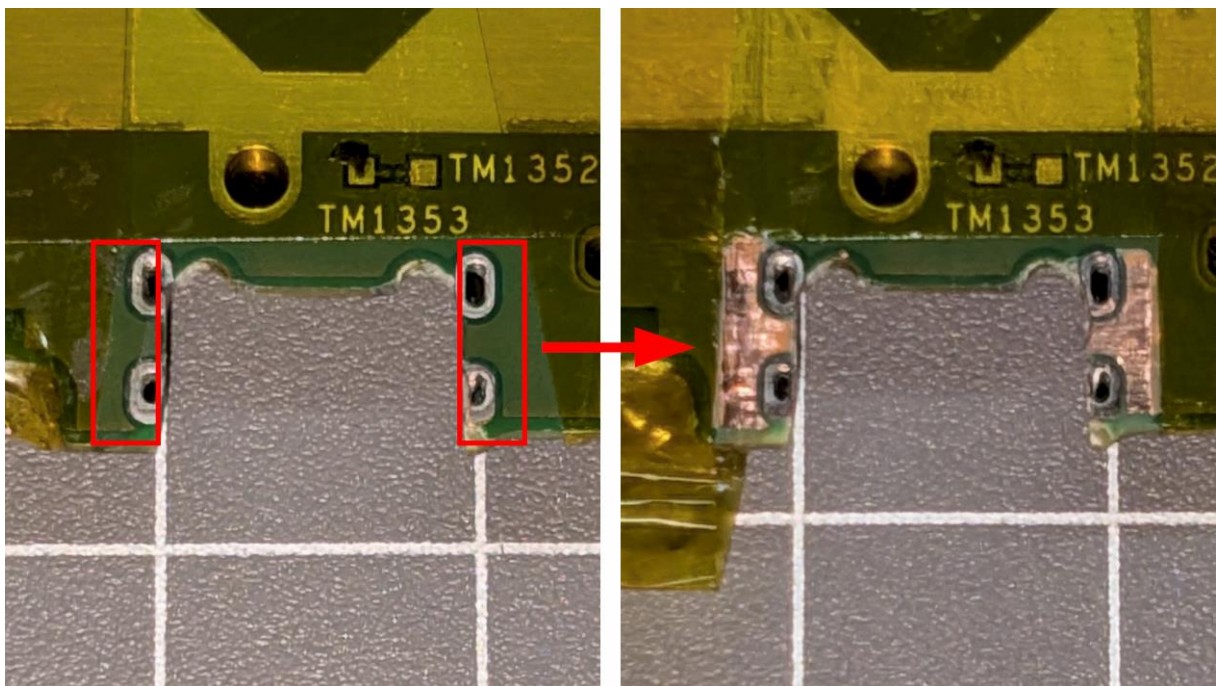


3. *PREPARE THE PADS FOR THE NEW CONNECTOR*

It is necessary to use the ground plane (GND) of the mainboard to firmly solder the shield connector. The ground plane is covered by the green soldermask. You need to use a cutter to scrape the surface and remove the paint that hides the copper.

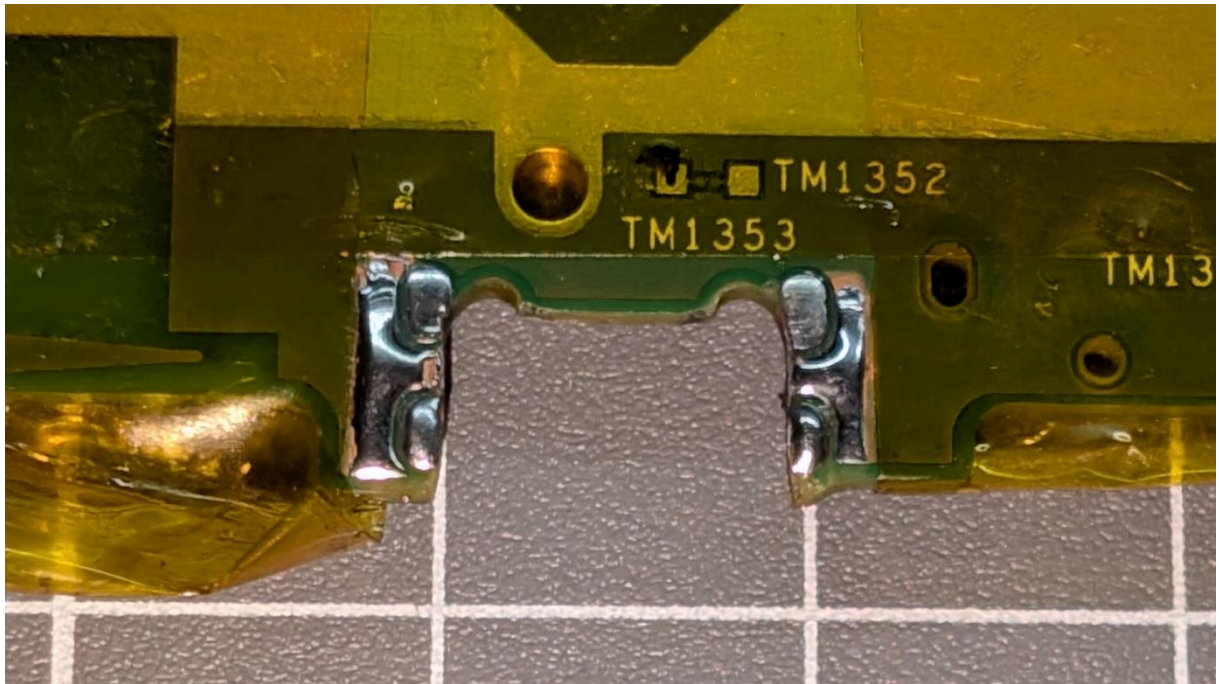
Once the copper is exposed, it will be possible to solder the connector in place.

With this image, the area where the green soldermask needs to be removed is marked in red color.



Once the copper is exposed, use solder and the soldering iron to cover it with solder. Also, fill the four pads where the original connector was soldered.

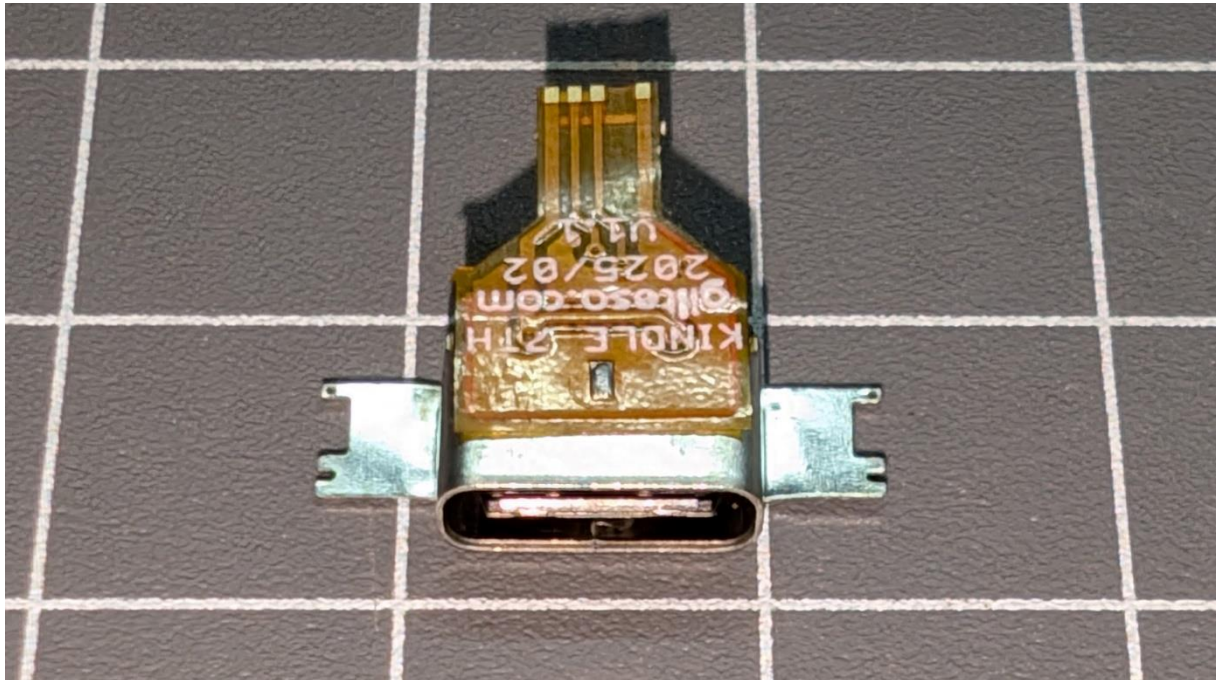
Don't use too much solder; just cover the area.



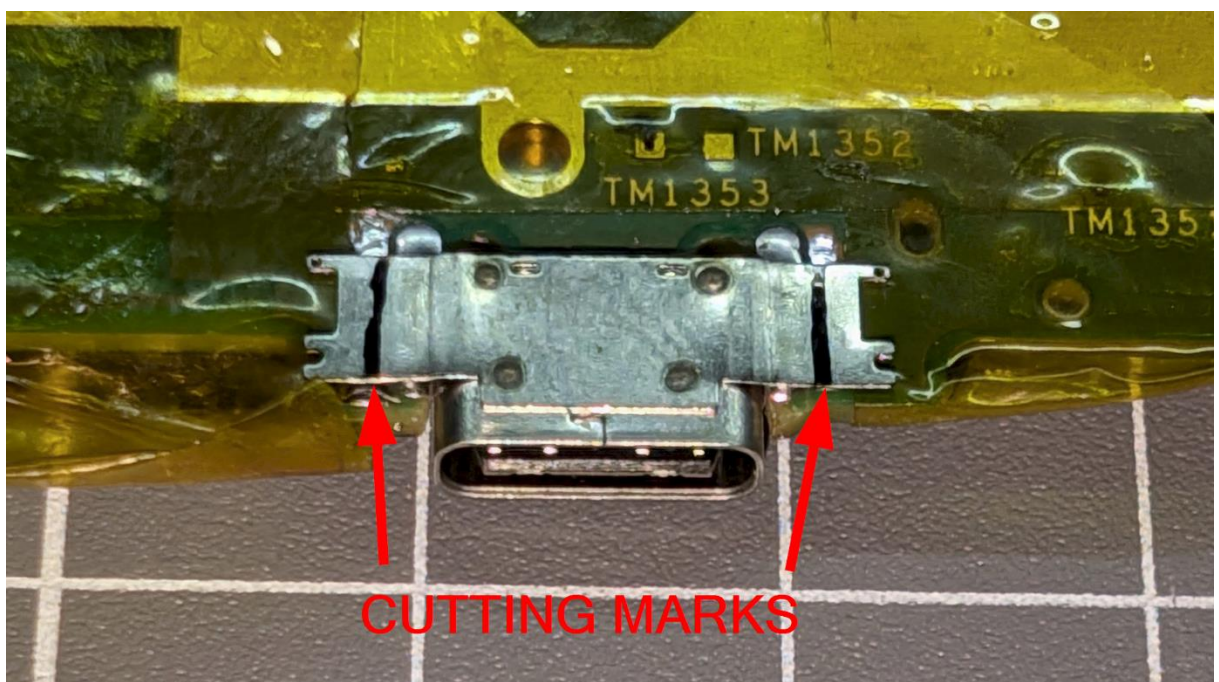
4. MODIFY THE USB-C CONNECTOR

To solder the USB-C connector, it will need to be modified first. The connector consists of two metal pieces joined together; on one side is the USB-C connector, and on the other, a metal plate that allows it to be soldered to the circuit.

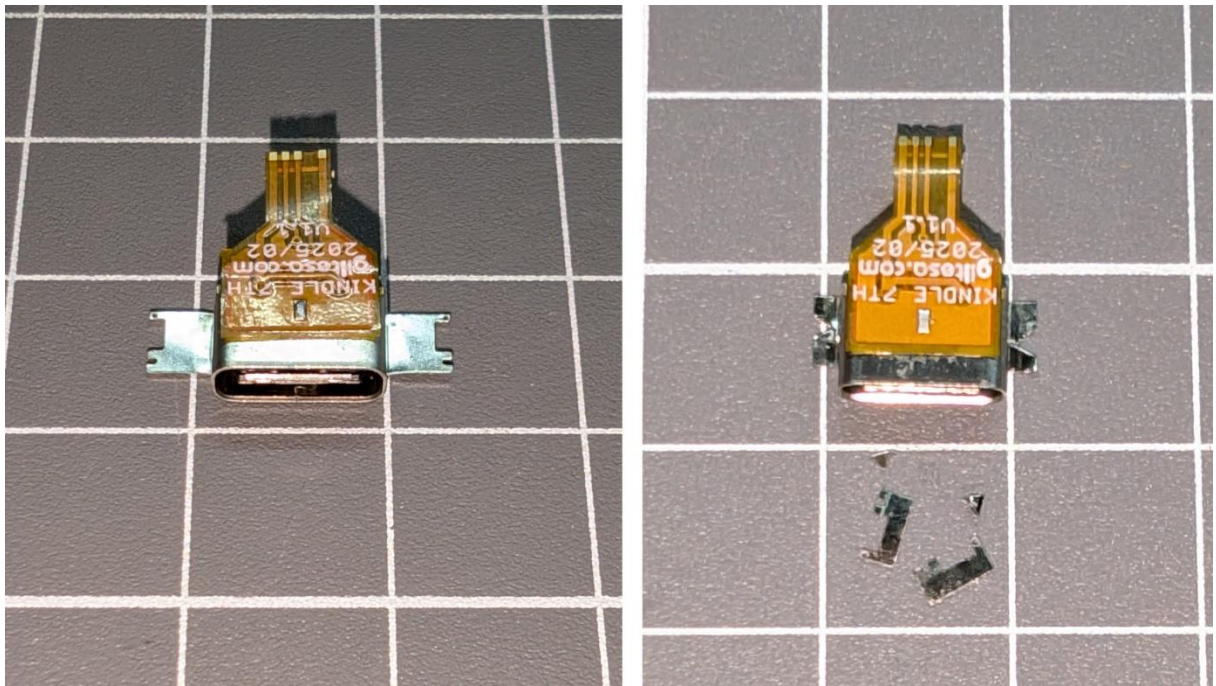
You will need to use a flat screwdriver to open the sides of the plate and then pliers to flatten it so that the USB-C connector has a pair of tabs.



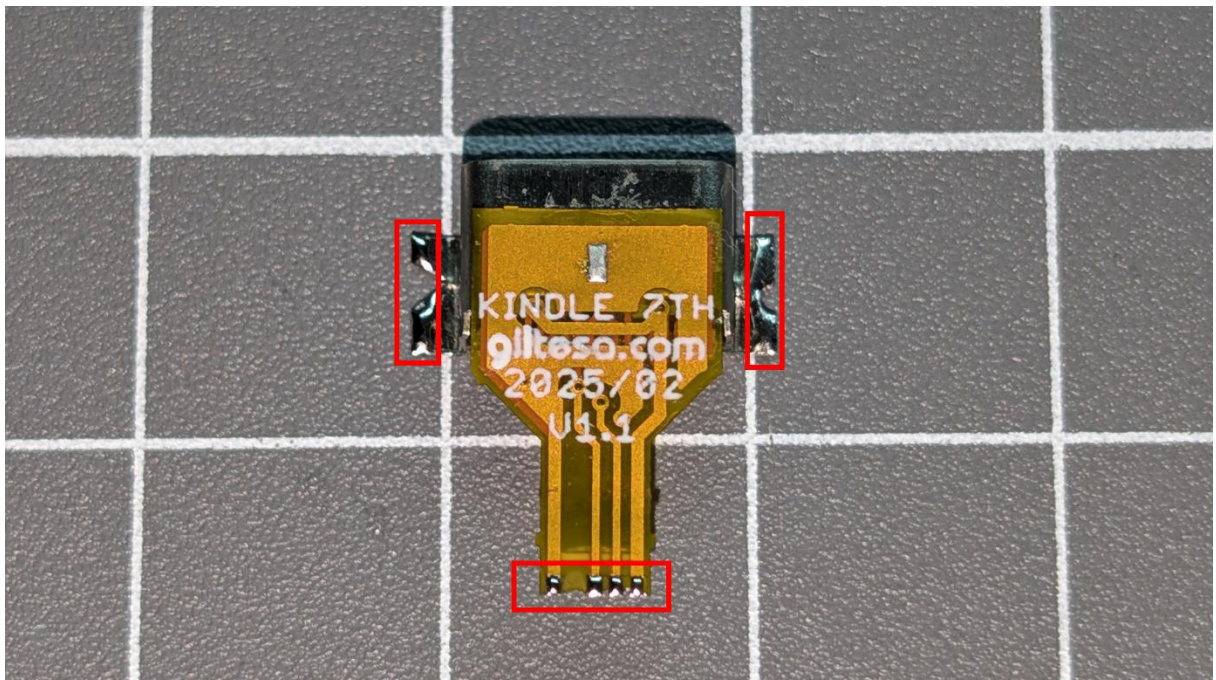
Next, you need to trim the length of these tabs to match your device.



To further improve the connection between the USB-C connector and the mainboard, you can make a V-shaped cut. Both modifications can be made with scissors.

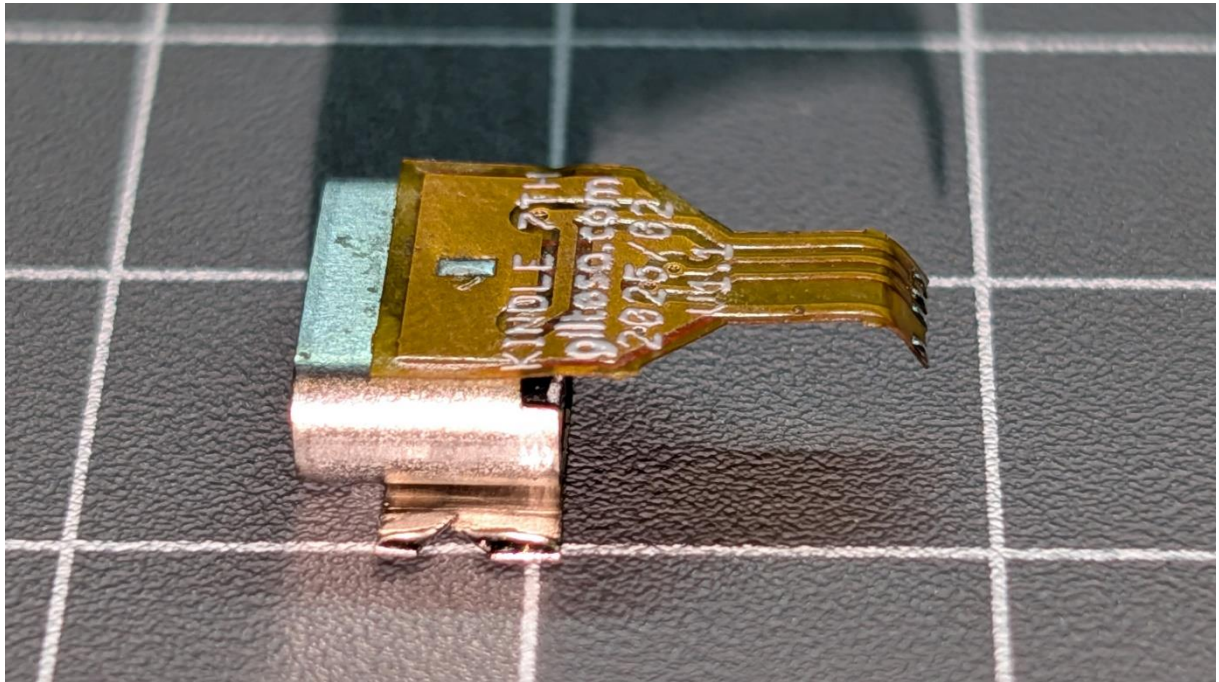


Once modified, take the opportunity to pre-solder both the newly cut metal plate, and the pads on the flexible circuit on both sides.

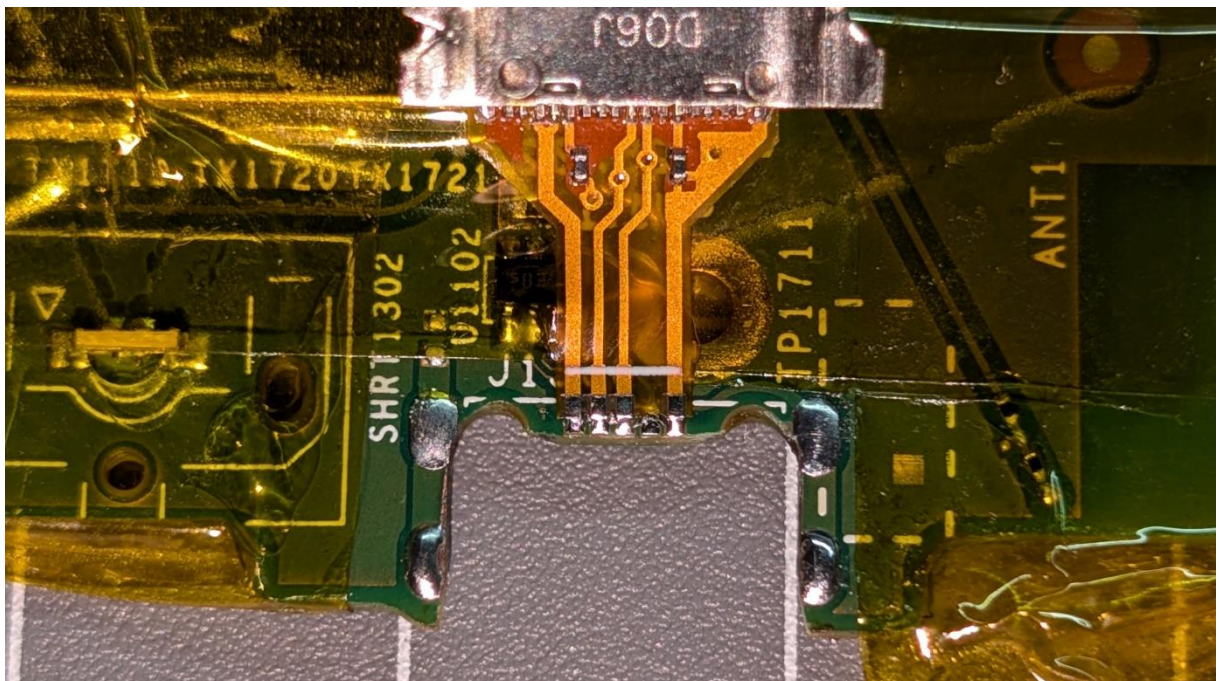


5. SOLDER THE BOARD TO THE MAINBOARD

Before soldering the board to the mainboard, slightly bend the flexible circuit like this:



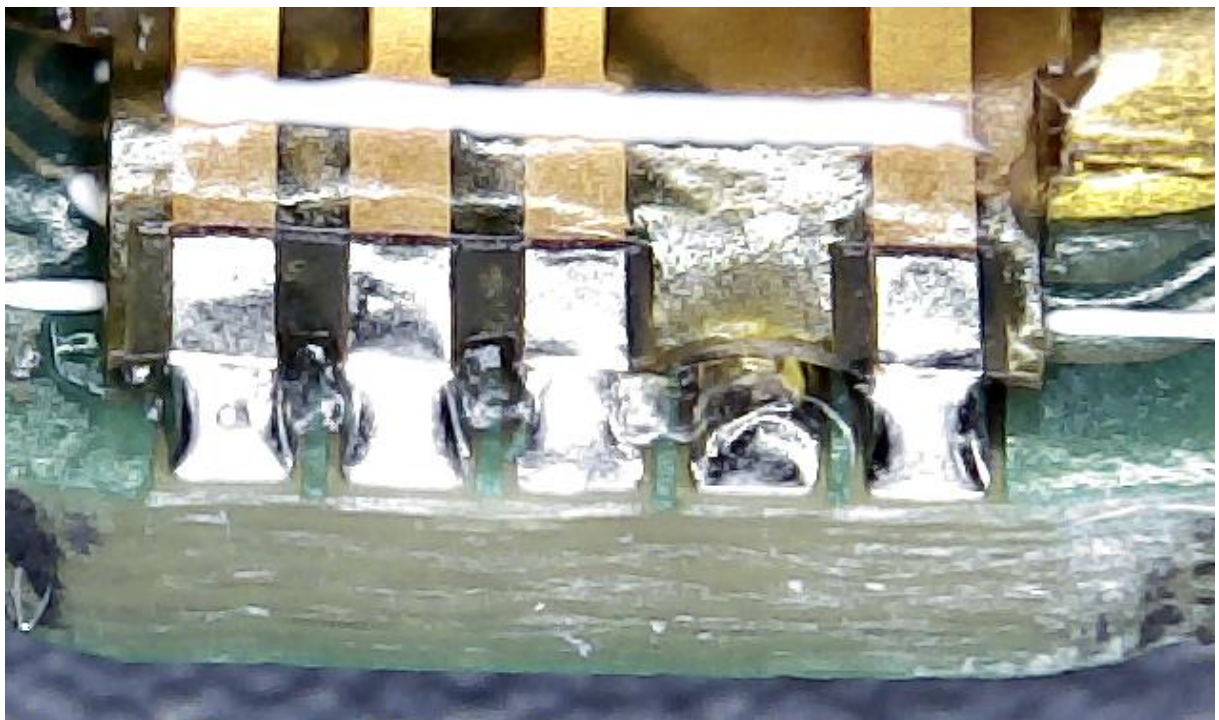
Now proceed to solder it, and make sure each pad is properly connected.



6. CONNECTION TEST

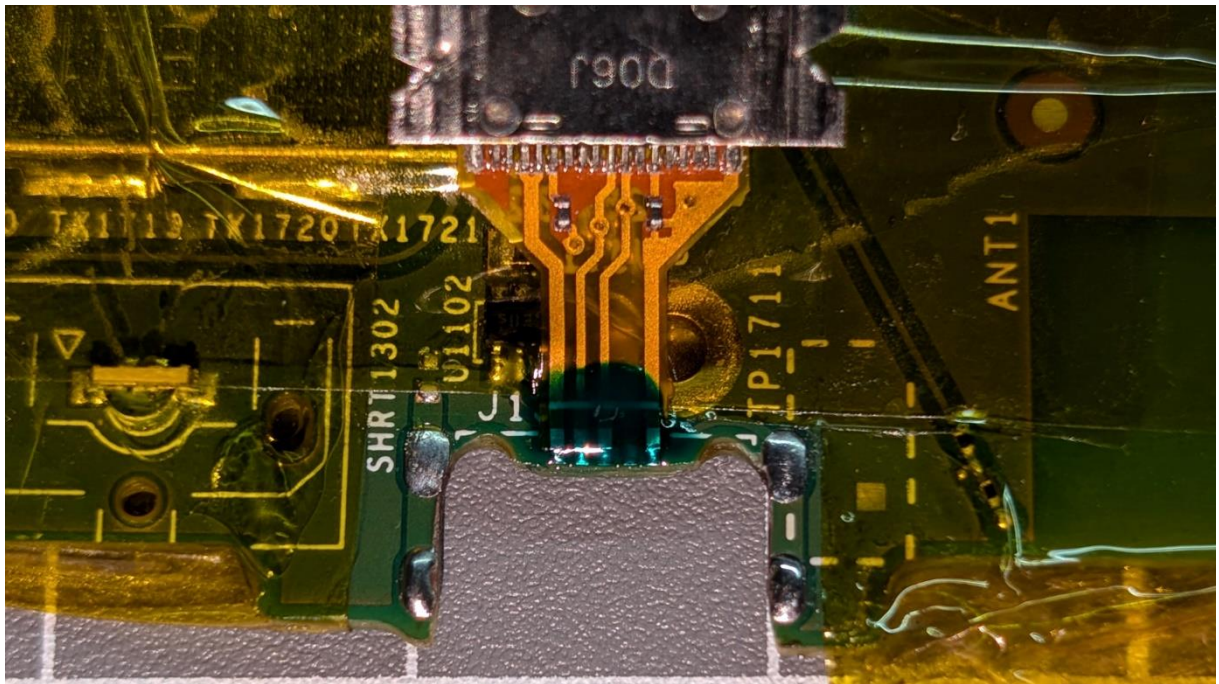
It is highly recommended to test now that the USB-C is working correctly with the device. To do this, you should connect the mainboard to the screen and the battery. Then, check that the battery charges when connected to the charger and that data can be transferred when connecting it to the computer.

If everything works fine, proceed with the installation. Otherwise, check that the solder joints are properly made. If you have a microscope or a camera/phone with a good camera zoom, use it to inspect the pads in detail.



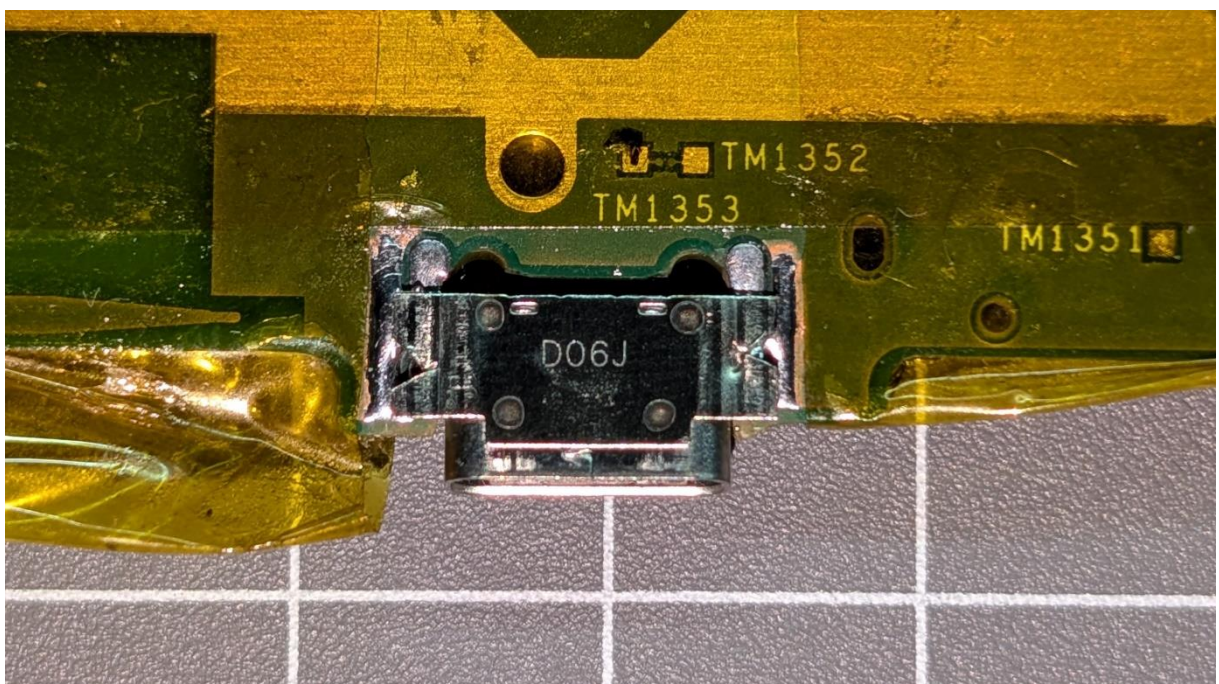
7. PROTECT THE PADS ON THE BOARD

It's advisable to cover the pads with kapton tape or solder mask to prevent any short circuits with the USB-C connector.

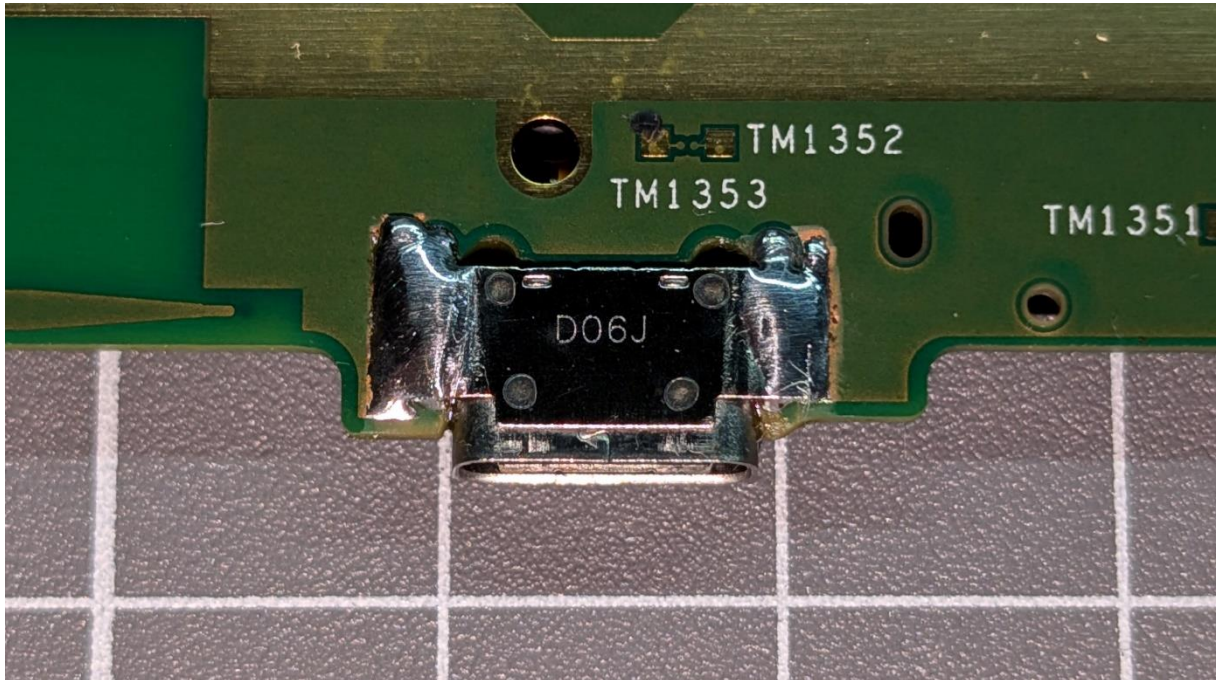


8. SOLDER THE USB-C SHIELD TO THE MAINBOARD

The final step in the USB-C installation is to solder the connector's shield to the mainboard. In the following image, the connector was not yet in its final position—**make sure it is fully inserted**, with the back of the connector touching the mainboard.



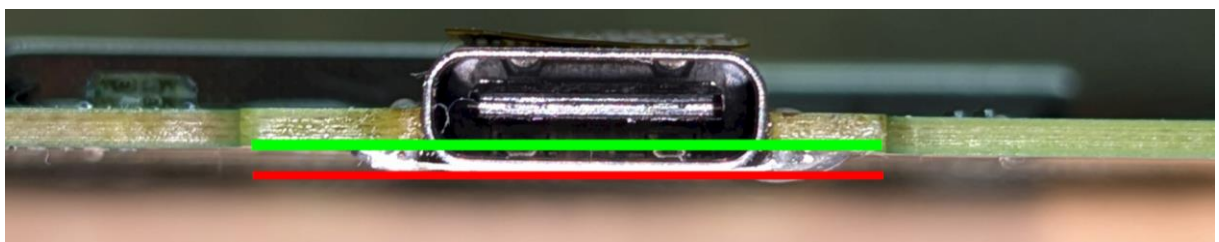
Here you can see how the connector looks once it's soldered.



You must ensure that the connector is not flush with the mainboard, but rather slightly positioned lower than the mainboard.



In this image, you can see how the connector is almost 1mm lower than the mainboard.



9. CUTTING THE PLASTIC SHELL

Assemble the device as if the installation were complete. You will notice that the USB-C cable cannot be connected because the new connector is wider and slightly taller than the MicroUSB. It's necessary to enlarge the hole in the casing. Use metal files or a utility knife to carefully and gradually enlarge the hole until you can connect the USB-C cable.



10. DONE!

The installation is complete for the Amazon Kindle Paperwhite 3. Follow the steps in reverse to close your device and enjoy it powered by USB-C!

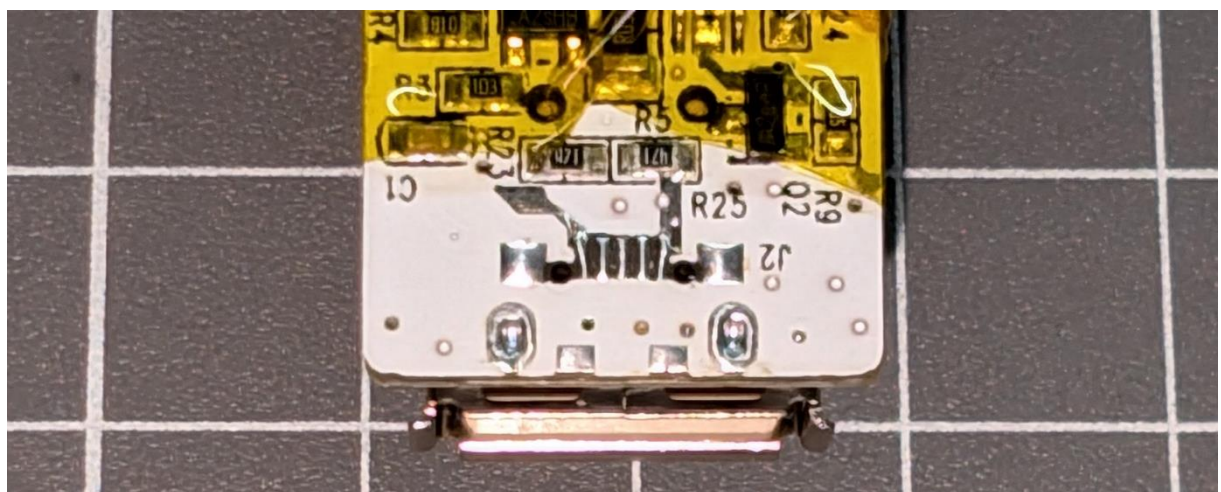
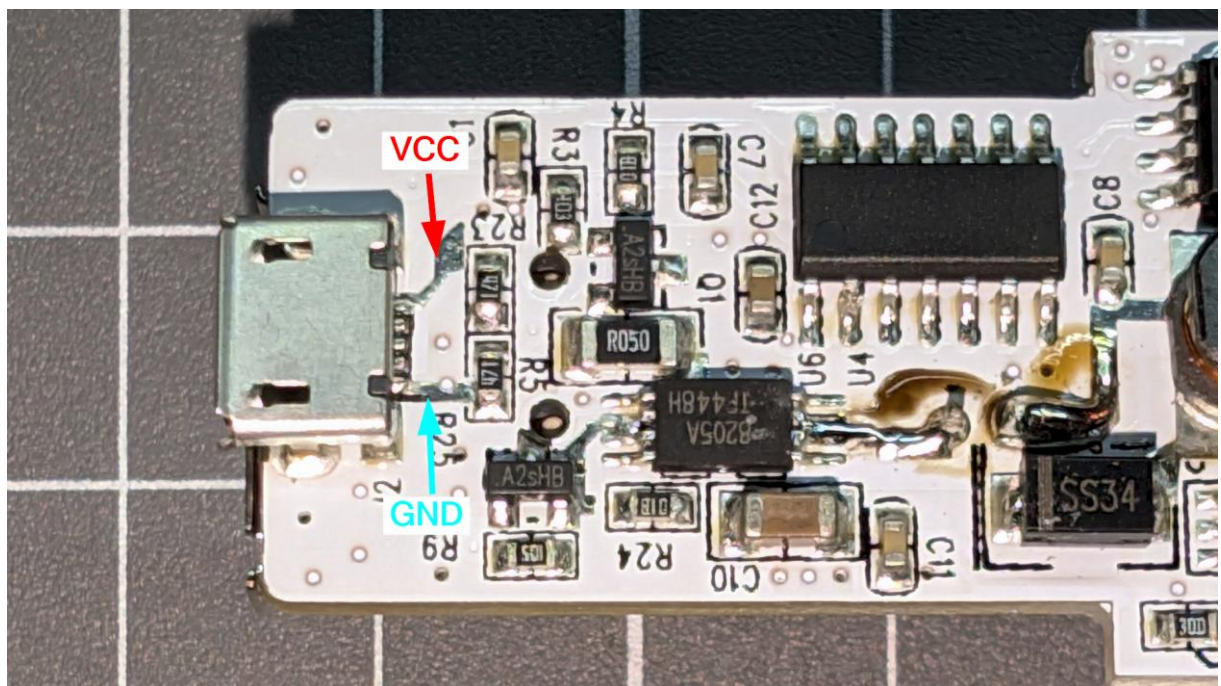


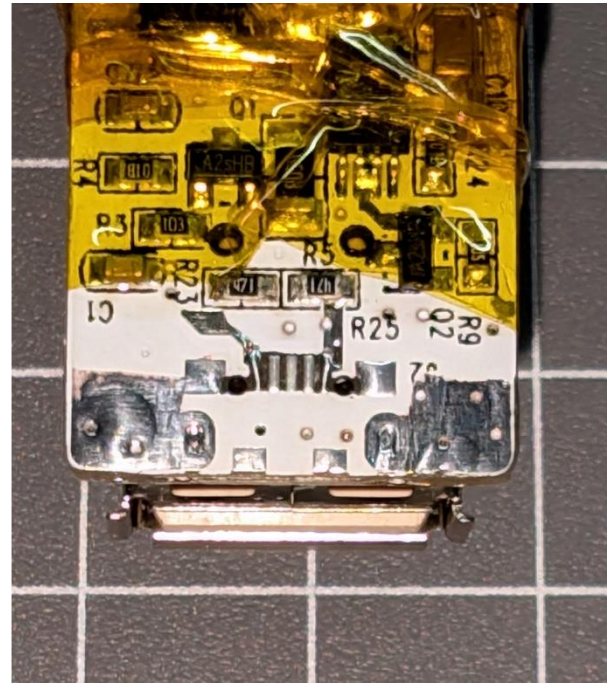
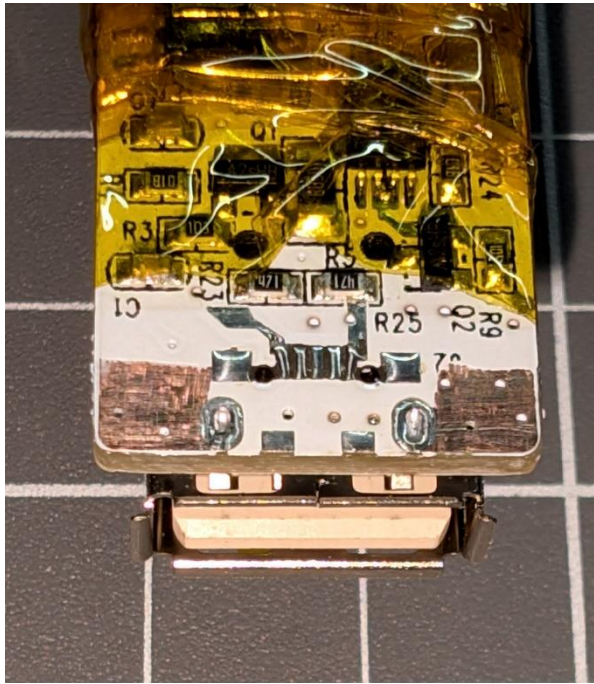
INSTALLATION ON BLITZWOLF BW-LT5 PRO [MICRO USB]

The installation is very similar for all the steps. The only difference in this device is that the MicroUSB used is soldered at its bottom to the mainboard through the shield's legs (instead of having a slot and being soldered to the sides).

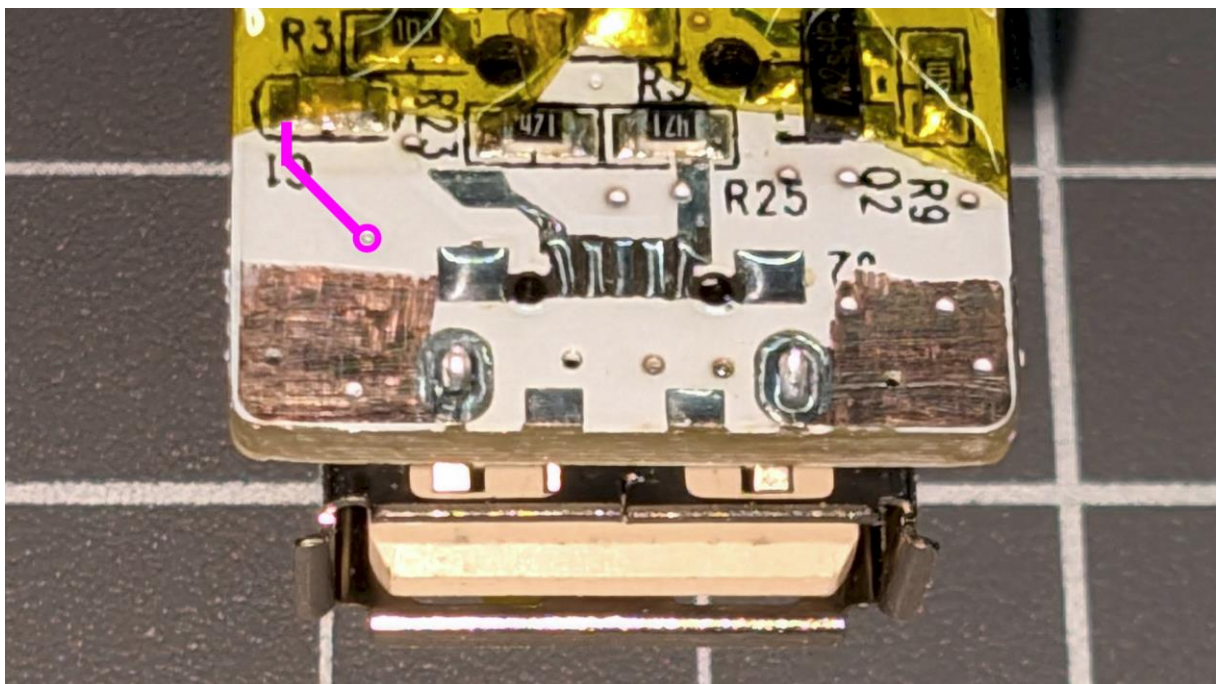
However, this does not change the installation process, as it is still necessary to modify the USB-C connector in the same way to create the two side tabs that will allow the connector to be soldered to the ground plane of the mainboard.

To avoid repeating what has already been described, the photos will be included without additional duplicated comments.

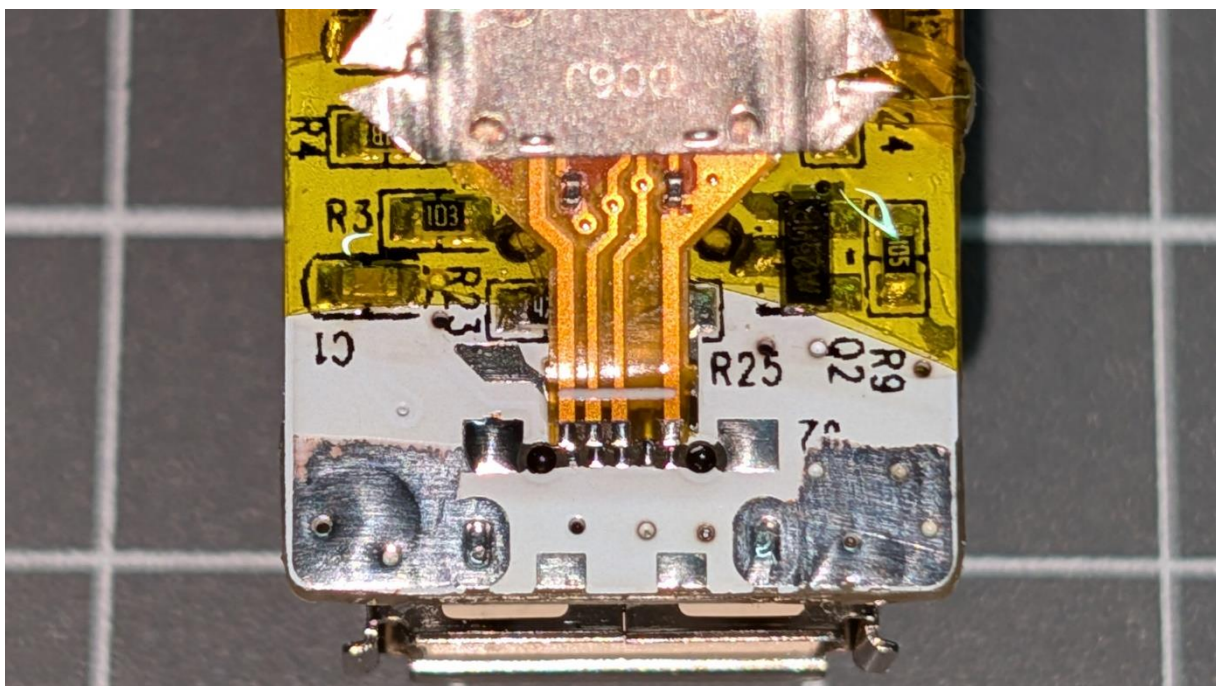
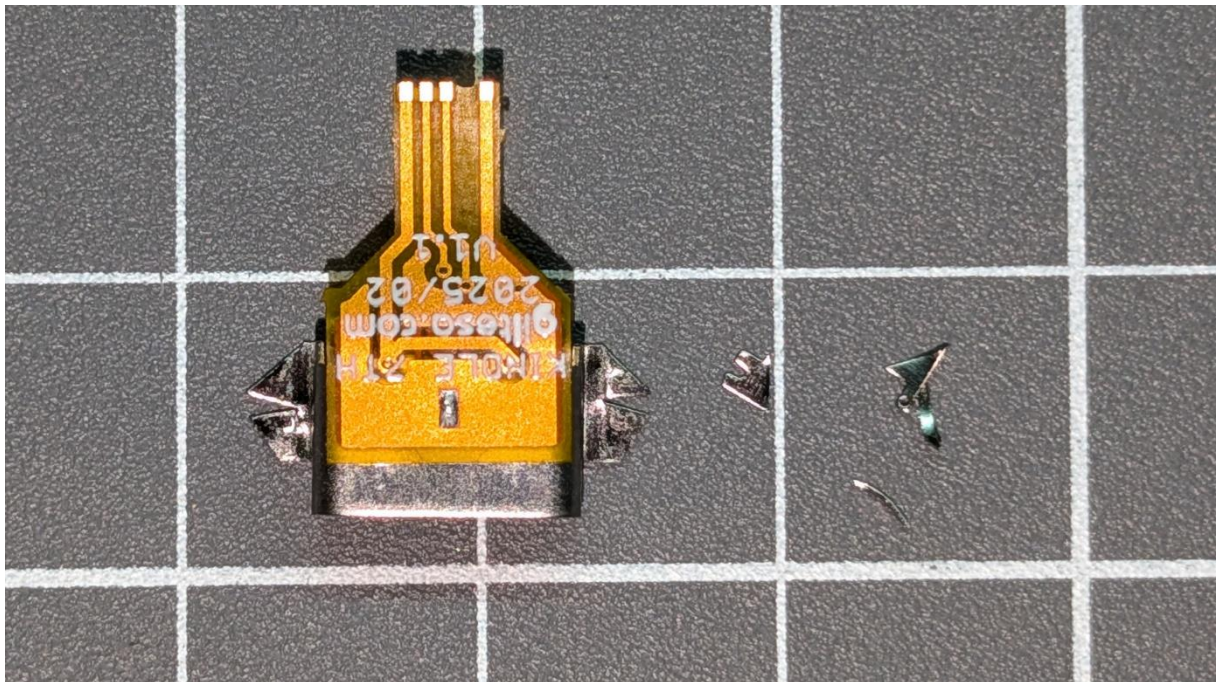


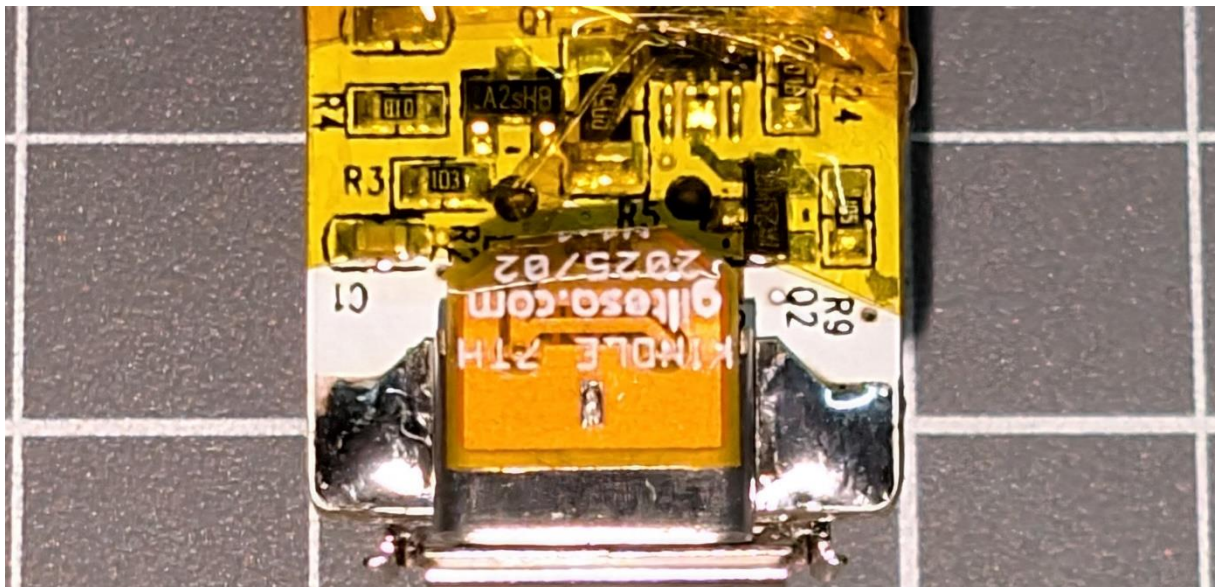
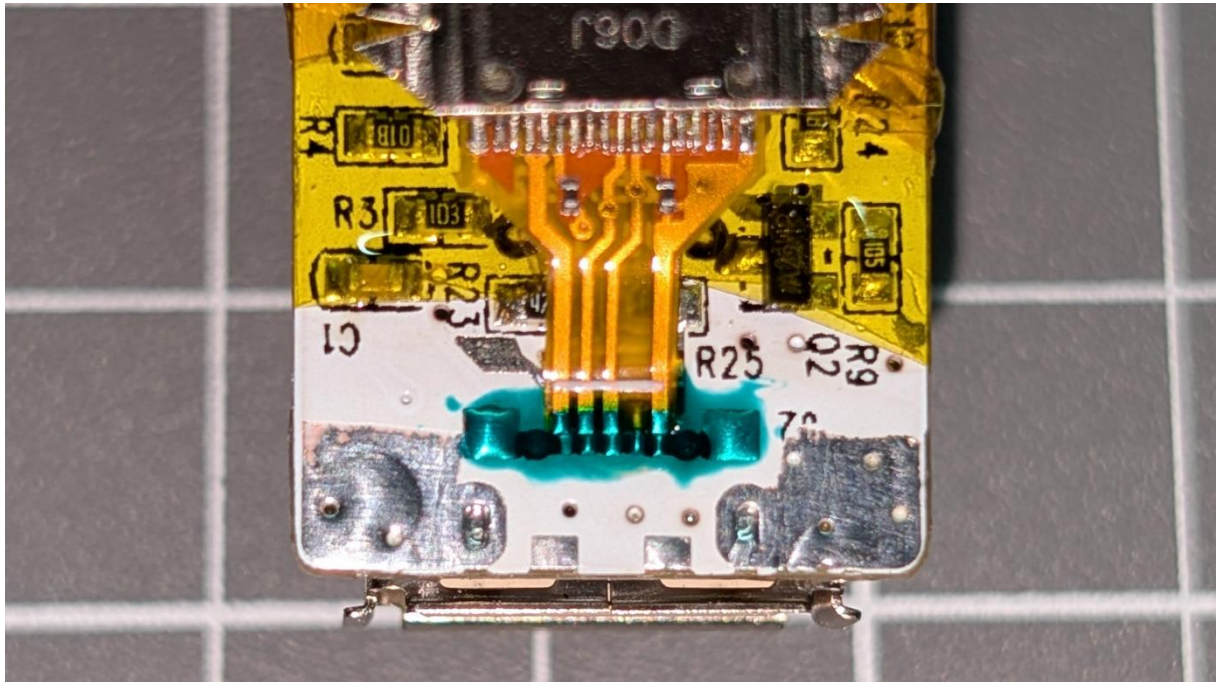


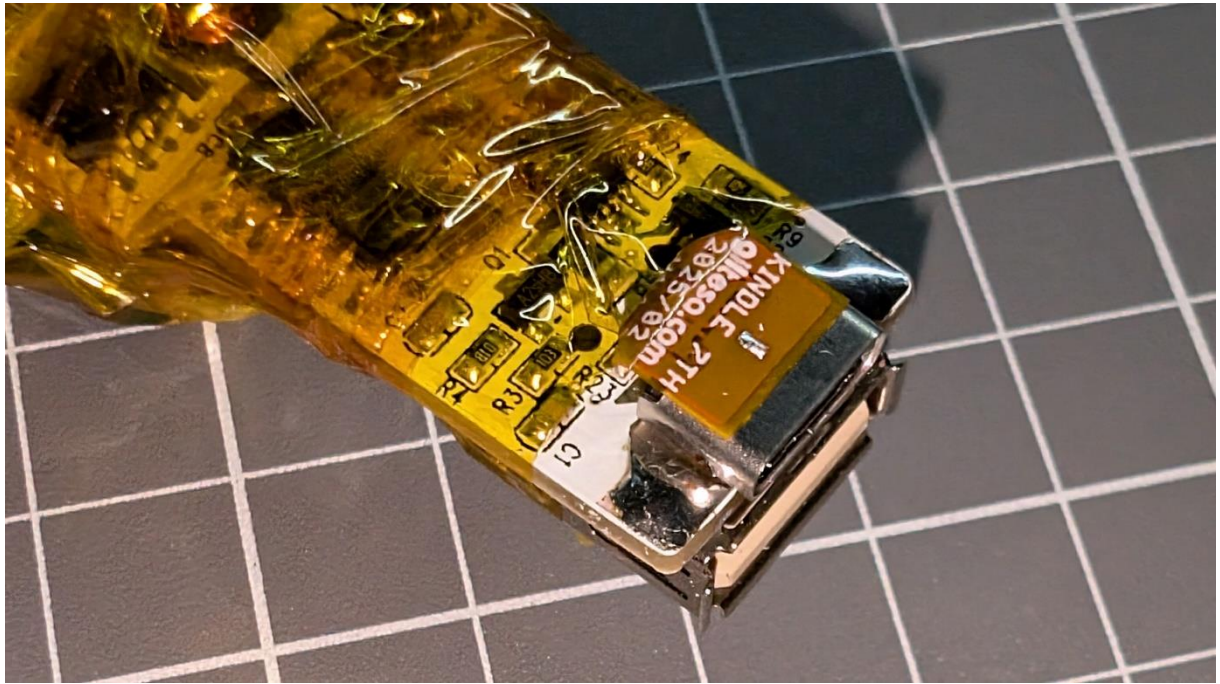
When removing the solder mask, be careful not to do so over any copper traces. In this image, there is one nearby, but it didn't interfere with the installation.



It's necessary to cut the tabs; in this case, they have a double V-shaped cut.







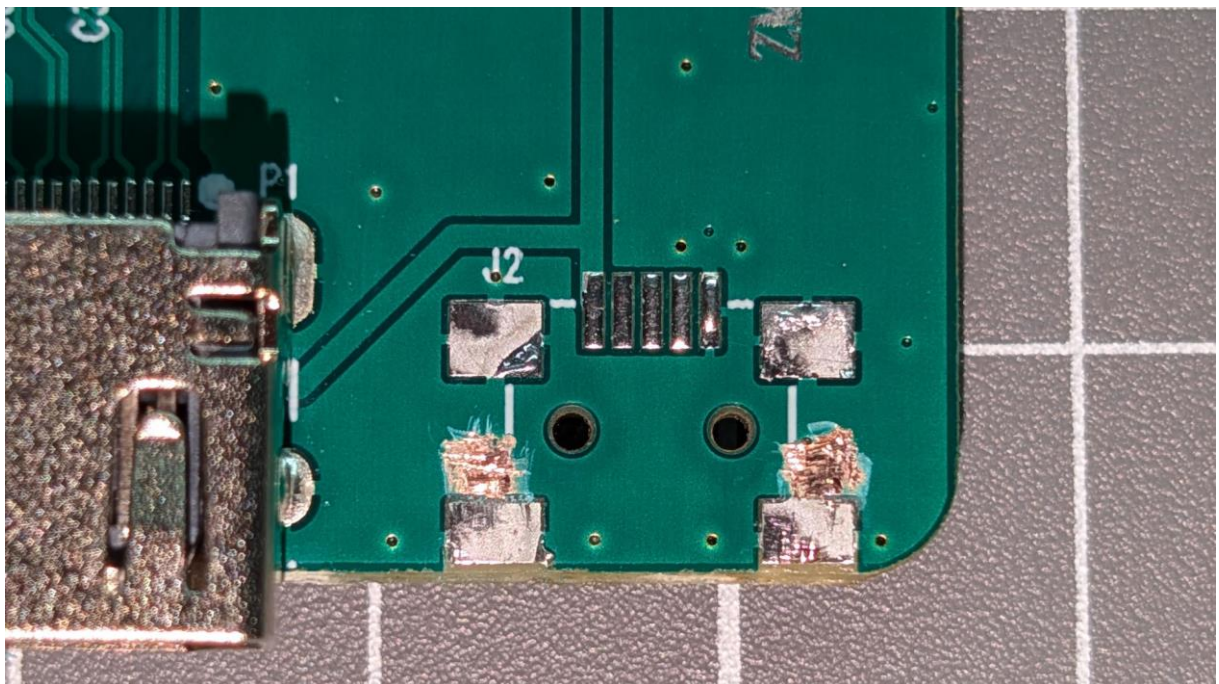
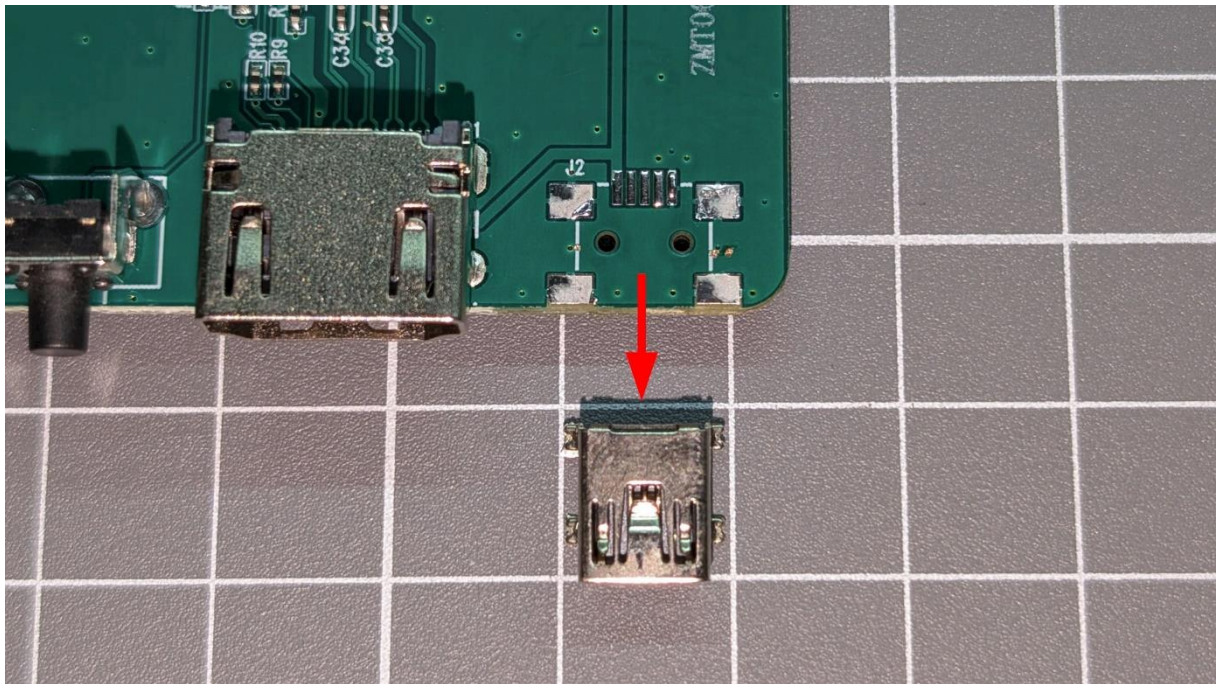


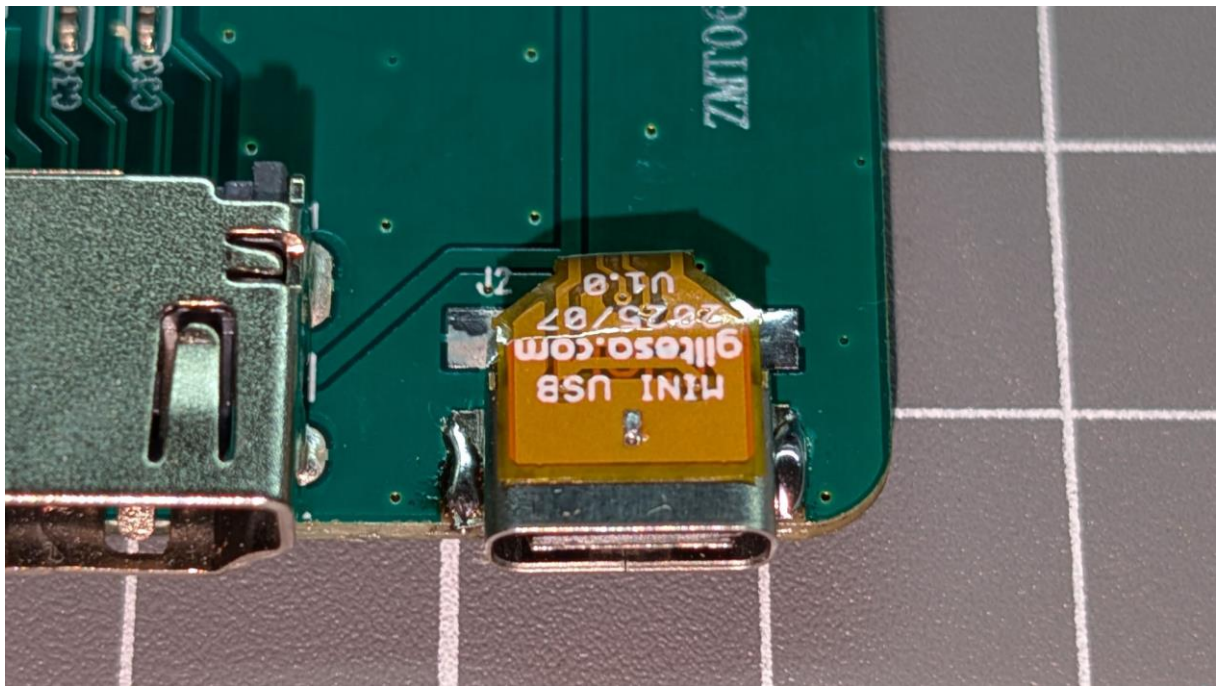
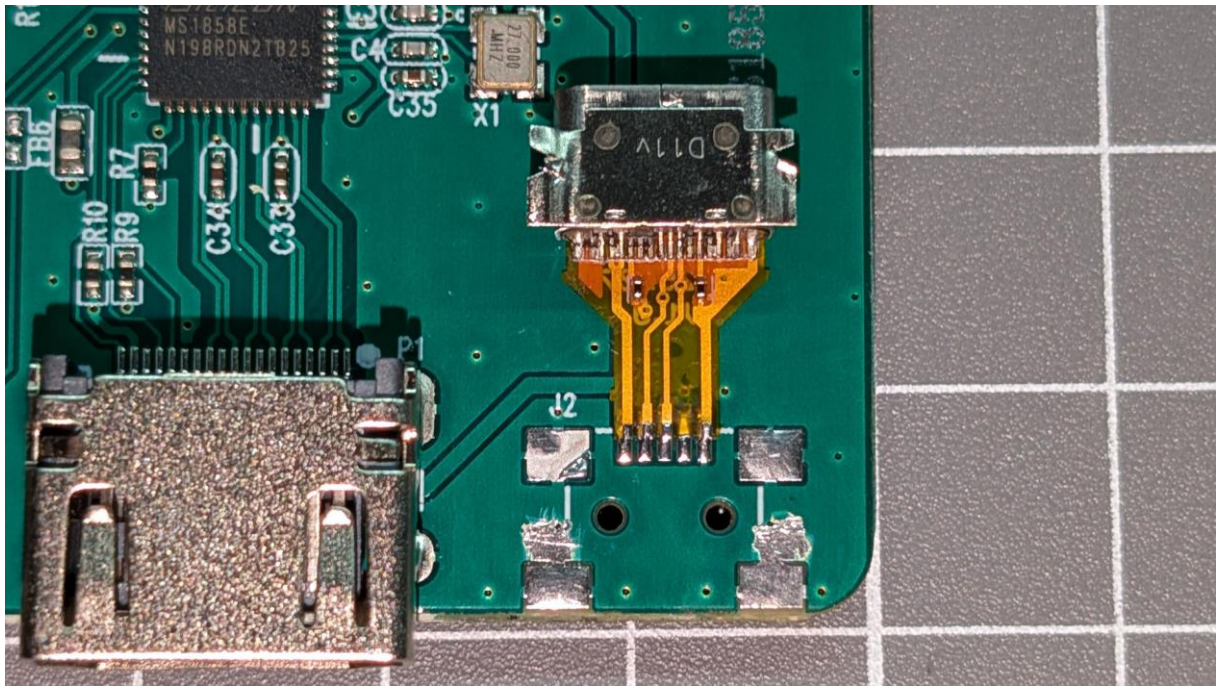
INSTALLATION ON RCA ADAPTER TO HDMI [MINI USB]

Another example, in this case on a cheap RCA to HDMI converter. It is powered by standard MiniUSB, so it is compatible.

As in previous examples, we must remove the connector, enlarge the pads where the USB will be soldered by scraping the solder mask and exposing the ground plane. Finally, solder the flexible cable and the USB-C shield to the board.

Installation photos:





FREQUENTLY ASKED QUESTIONS - FAQ

WHAT CHARGER CAN BE USED?

You can use any standard charger for mobile phones, computers, etc., with 5V 1A. It doesn't need to be a Power Delivery charger since this feature is not used. Of course, if you want to use a Power Delivery charger, there's no problem or risk.

Technical data for curious minds:

Power Delivery chargers can supply a wide range of voltages: 5V, 9V, 12V, 15V, and 20V. However, for this to happen, the device must communicate with the charger to explicitly request the desired voltage. Without this communication, the charger will never supply more than 5V. That's one of the advantages of USB-C, as it can be used with both old and modern devices.

CAN I CONNECT THE DEVICE TO THE PC?

Of course, whether it can transfer data or just charge will depend on the functions of your device. In the case of the Amazon Kindle, you will be able to transfer books or any other data, just as you could before with the MicroUSB. In the case of the flashlight, the connector was used only for charging the device, so with the new USB-C, it will serve the same purpose.